

Database development with PL/SQL INSY 8311



ADVENTIST UNIVERSITY
OF CENTRAL AFRICA

Instructor:

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6h00 pm – 8h50 pm

- Monday A -G209
- Tuesday B-G209
- Wednesday E-G209
- Thursday F-G308

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Database development with PL/SQL

Reference reading



- [4 PL/SQL Control Statements](#)
- [Oracle PL/SQL Insert, Update, Delete & Select Into \[Example\]](#)
- [YouTube Tutorial: Oracle CDBs and PDBs Explained \(With Connections\)](#)
- [Oracle 12c: Applying PSU 12.1.0.1.1 with Multitenant DB & unplug/plug](#)
- [Multitenant : Overview of Container Databases \(CDB\) and Pluggable Databases \(PDB\)](#)
- [1 Introduction to the Multitenant Architecture](#)

Lecture 05 – Oracle Database Environment (CDBs & PDBs) & OEM

Course Objectives: CDB, PDB, and Oracle Enterprise Manager



Understand Container Database (CDB) and Pluggable Database (PDB) architecture and their role in Oracle multi-tenancy.

Create, configure, and manage CDBs and PDBs using SQL*Plus and Oracle tools.

Perform SQL operations within PDBs, including creating schema objects.

Administer databases using Oracle Enterprise Manager (OEM) for monitoring and management.

Monitor and optimize database performance using OEM tools.

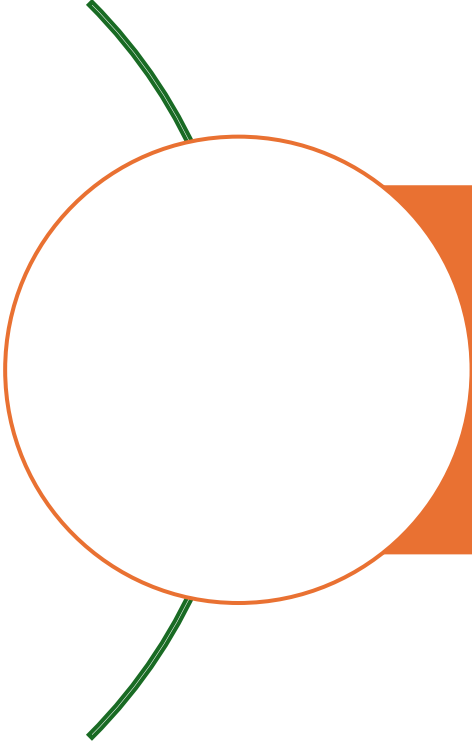
Automate routine tasks and schedule jobs via OEM.

Manage users and security in CDB and PDB environments.

Execute backup and recovery operations at both CDB and PDB levels.

Troubleshoot common database issues using OEM diagnostic tools.

Oracle Databases CDBs and PDBs

A decorative graphic consisting of a white circle with an orange outline, positioned to the left of the text box. Two green curved lines extend from the top and bottom of the circle.

In Oracle databases, **CDBs (Container Databases)** and **PDBs (Pluggable Databases)** are part of Oracle's Multitenant Architecture, introduced in Oracle Database 12c. This architecture allows for better resource management, scalability, and easier database consolidation.

Multitenant Container Database Architecture

Oracle's **Multitenant Architecture** enables efficient management of multiple databases through a single **Container Database (CDB)**. Here's a simplified explanation:

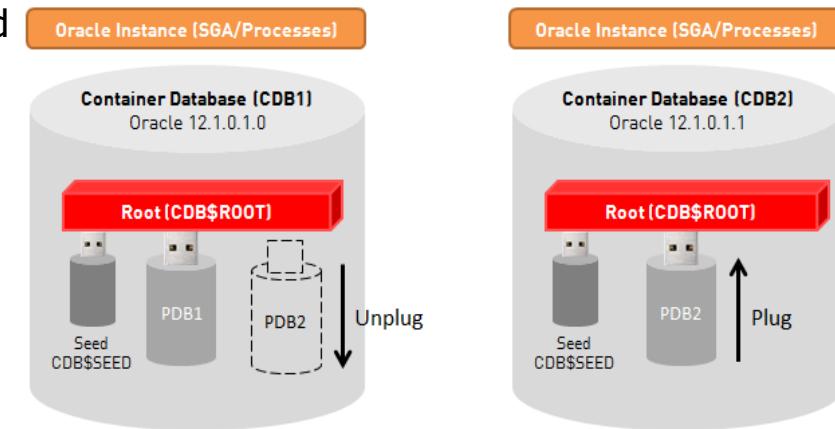


1. Container Database (CDB)

- A **CDB** is the main database that contains all necessary components to manage multiple **Pluggable Databases (PDBs)**.
- The **Root Container (CDB\$ROOT)** holds system-wide metadata and common objects shared by all PDBs. However, it does not store user data.
- The **PDB\$SEED** is a template used to quickly create new PDBs, making it easy to spin up new databases.

2. Pluggable Databases (PDBs)

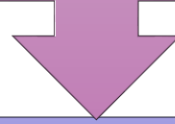
- **PDBs** are independent databases within the CDB that contain their own data, tablespaces, schemas, and administrative settings.
- Each PDB functions like a standalone database but shares resources (CPU, memory, storage) with other PDBs in the same CDB.
- PDBs can be unplugged from one CDB and plugged into another, providing flexibility for database movement.



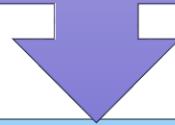
How an Instance Works



When an Oracle instance is started, it allocates the necessary memory structures Program Global Area(PGA) and System Global Area (SGA) and starts the associated background processes.



Once started, users can connect to the instance to perform operations like querying data, modifying tables, and running applications.



When the instance is shut down, memory is released, and the background processes are terminated.

Advantages of Multitenant Architecture & Use Cases

Advantages of Multitenant Architecture

1. Efficient Resource Use

- PDBs within a CDB share system resources, reducing hardware costs while maintaining database isolation.

2. Database Consolidation

- Multiple databases (PDBs) can exist within one CDB, simplifying management.

3. Simplified Maintenance

- Upgrades and patches applied to the CDB are automatically applied to all PDBs, streamlining maintenance.

4. Portability

- PDBs can be easily moved between CDBs, which is particularly useful in cloud environments.



Summary

Oracle's multitenant architecture allows you to manage multiple databases **more efficiently, providing flexibility and ease of use**, while keeping each PDB isolated and secure.

Use Cases

Multitenancy

- Allows cloud providers to host multiple customer databases in a single CDB without data intermingling.

Development and Testing

- Developers can use isolated PDBs on a shared server, working independently.

CDBs and PDBs: part of Oracle's Multitenant Architecture



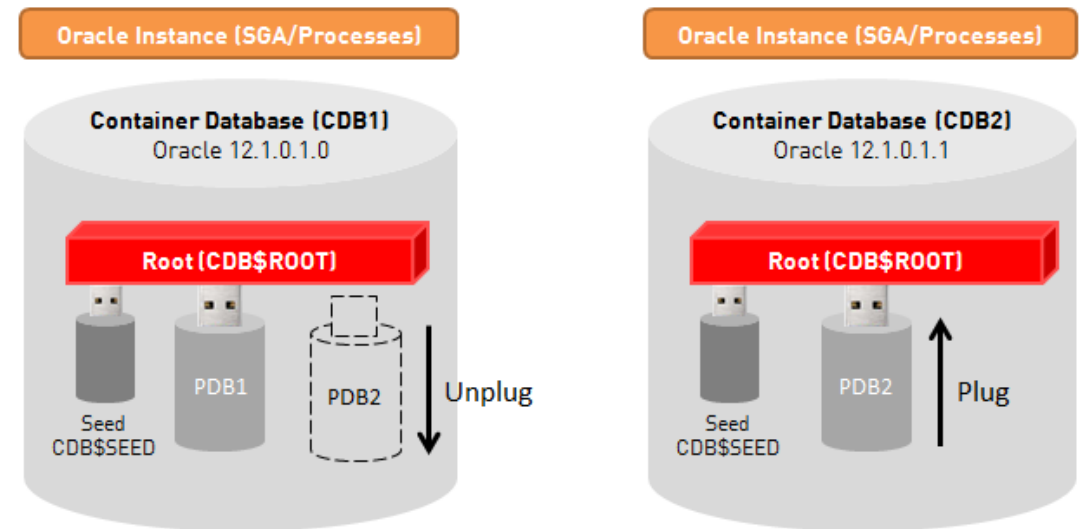
Explanation of the Image:

1. Left Side (CDB1 - Unplugging):

- In **CDB1**, we see two pluggable databases:
 - **PDB1**
 - **PDB2** (which is being unplugged)
- There's also a **seed database (CDB\$SEED)**, which is a template used to create new PDBs.
- The process shown here is **unplugging** of **PDB2** from **CDB1**, meaning that PDB2 is removed from CDB1 but remains available to be plugged into another container database (CDB2).

2. Right Side (CDB2 - Plugging):

- In **CDB2**, **PDB2** is being **plugged** into this new container database.
- After unplugging from CDB1, PDB2 can be moved and plugged into another CDB, in this case, **CDB2**.
- As in CDB1, CDB2 also has a **seed database (CDB\$SEED)** and its own instance (SGA/Processes).



Source Image: <https://www.dbi-services.com/blog/oracle-12c-applying-psu-121011-with-multitenant-db-and-unplugplug/>

Display all PDBS

```
Command Prompt - sqlplus s X + v
Microsoft Windows [Version 10.0.22631.4037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Eric>sqlplus sys as SYSDBA

SQL*Plus: Release 21.0.0.0.0 - Production on Sun Sep 22 14:36:16 2024
Version 21.3.0.0.0

Copyright (c) 1982, 2021, Oracle. All rights reserved.

Enter password:

Connected to:
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

SQL> show user;
USER is "SYS"
SQL>
SQL> SELECT instance_name FROM v$instance;

INSTANCE_NAME
-----
oracle21cdb

SQL> show pdbs;

  CON_ID CON_NAME              OPEN MODE RESTRICTED
-----
2 PDB$SEED                READ ONLY NO
3 ORACLE21DBCPDB          READ WRITE NO

SQL> |
```

- To connect to an Oracle database using SQL*Plus as the SYS user with SYSDBA privileges.

SQL> sqlplus sys as SYSDBA



- Oracle instance that is currently running, which is usually the same as the **SID** (System Identifier) of the database.

SQL> SHQW PDBS;

- This PDB helps streamline the process of managing multiple PDBs within a container database (CDB).

- This is the PDB created during the installation of Oracle 21c; for others, you might have XE.

Show path of PDBs

```
SQL> show pdbs;
```

CON_ID	CON_NAME	OPEN MODE	RESTRICTED
2	PDB\$SEED	READ ONLY	NO
3	ORACLE21DBCPDB	READ WRITE	NO

```
SQL> SELECT CON_ID, TABLESPACE_NAME, FILE_NAME
2 FROM CDB_DATA_FILES
3 WHERE CON_ID = 3;
```

CON_ID	TABLESPACE_NAME	FILE_NAME
3	SYSTEM	E:\ORACLE21CDATA\ORACLE21CDB\ORACLE21DBCPDB\SYSTEM01.DBF

3	SYSAUX	E:\ORACLE21CDATA\ORACLE21CDB\ORACLE21DBCPDB\SYSAUX01.DBF
---	--------	--

3	UNDOTBS1	E:\ORACLE21CDATA\ORACLE21CDB\ORACLE21DBCPDB\UNDOTBS01.DBF
---	----------	---

CON_ID	TABLESPACE_NAME	FILE_NAME
3	USERS	E:\ORACLE21CDATA\ORACLE21CDB\ORACLE21DBCPDB\USERS01.DBF

```
SQL> |
```

- This query retrieves the tablespace names and file names of data files for the Pluggable Database (PDB) with CON_ID = 3 from the CDB_DATA_FILES view in an Oracle multitenant environment. - **Location of XE PDB1**

```
SQL> SELECT CON_ID, TABLESPACE_NAME, FILE_NAME FROM
CDB_DATA_FILES WHERE CON_ID = 3;
```

- This path is commonly used for organizing database files, making it easier to manage multiple PDBs within a CDB.

Create PDB using PDBSEED

```
Command Prompt - sqlplus s X + v
SQL*Plus: Release 21.0.0.0.0 - Production on Sun Sep 22 17:30:49 2024
Version 21.3.0.0.0

Copyright (c) 1982, 2021, Oracle. All rights reserved.

Enter password:

Connected to:
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

SQL> CREATE PLUGGABLE DATABASE PDB2
  2 ADMIN USER pdbadmin IDENTIFIED BY admin
  3 FILE_NAME_CONVERT = ('E:\oracle21cdata\ORACLE21CDB\pdbseed\',
  4 'E:\oracle21cdata\ORACLE21CDB\oracle21dbcpdb\PDB2\');

Pluggable database created.

SQL>
SQL>
SQL> SELECT pdb_name, status FROM cdb_pdbs;

PDB_NAME
-----
STATUS
-----
ORACLE21DBCPDB
NORMAL

PDB$SEED
NORMAL

PDB2
NEW
```

- This creates a new pluggable database by copying the structure of pdbseed and setting up a new admin user.

```
SQL> CREATE PLUGGABLE DATABASE PDB2
ADMIN USER pdbadmin IDENTIFIED BY admin
FILE_NAME_CONVERT =
('E:\oracle21cdata\ORACLE21CDB\pdbseed\',
'E:\oracle21cdata\ORACLE21CDB\oracle21dbcpdb\PDB2\');
```

- It retrieves the name and status of all Pluggable Databases (PDBs) from the CDB_PDBS view in an Oracle multitenant environment.
- PDB2 has status new because it has not yet opened

```
SQL> SELECT pdb_name, status from cdb_pdbs
```

PDB Status

```
SQL> select name, open_mode from v$pdb;
```

NAME

OPEN_MODE

PDB\$SEED
READ ONLY
ORACLE21DBCPDB
READ WRITE
PDB2
MOUNTED

```
SQL> SELECT NAME, CON_ID FROM v$active_services order by 1;
```

NAME	CON_ID
-----	-----
SYS\$BACKGROUND	1
SYS\$USERS	1
oracle21cdb	1
oracle21cdbXDB	1
oracle21dbcpdb	3
pdb2	4

6 rows selected.

```
SQL> |
```

- In short, this query gives you the status of all PDBs in the CDB.
- The pdb created is mounted = not yet open for use.

```
SQL> select name, open_mode from v$pdb;
```

- The command lists active services and their container IDs (CDB or PDB), sorted by service name.

```
SQL> SELECT NAME, CON_ID FROM v$active_services ORDER BY 1;
```

Display Current Container – CDB\$ROOT

```
Command Prompt - sqlplus s X + v
SQL> ALTER PLUGGABLE DATABASE PDB2 OPEN;
Pluggable database altered.
SQL> SHOW CON_NAME;
CON_NAME
-----
CDB$ROOT
SQL>
SQL>
SQL> SELECT name, open_mode from v$pdb;
NAME
-----
OPEN_MODE
-----
PDB$SEED
READ ONLY
ORACLE21DBCPDB
READ WRITE
PDB2
READ WRITE
```

- This command will return the name of the current container you are connected to, which could be the root container (CDB\$ROOT) or a specific pluggable database (PDB).



SQL> Show con_name;
or
SQL> SELECT SYS_CONTEXT('USERENV', 'CON_NAME') FROM DUAL;

- Retrieves the names and open modes (like READ WRITE, READ ONLY) of all pluggable databases (PDBs) in the Oracle container.

SQL> SELECT name, open_mode from v\$pdb;

Alter PDB2 – Open & Save

```
SQL> ALTER PLUGGABLE DATABASE PDB2 OPEN;  
ALTER PLUGGABLE DATABASE PDB2 OPEN  
*  
ERROR at line 1:  
ORA-65019: pluggable database PDB2 already open  
  
SQL> -- Save the state so PDB2 opens automatically on CDB startup  
SQL> ALTER PLUGGABLE DATABASE PDB2 SAVE STATE;  
  
Pluggable database altered.  
  
SQL> |
```

- After creating the PDB, it will be in a mounted state. The command opens the **PDB2** pluggable database.

```
SQL> ALTER PLUGGABLE DATABASE PDB2 OPEN;
```

- The command saves PDB pdb2's state for automatic opening in the same mode (READ ONLY or READ WRITE) on the next CDB startup.

```
SQL> ALTER PLUGGABLE DATABASE pdb2 SAVE STATE;
```



Oracle Instance

```
SQL> SELECT instance_name FROM v$instance;
```

```
INSTANCE_NAME
```

```
-----  
oracle21cdb
```

- This Query retrieves the name of the current Oracle database instance you are connected to.

```
SQL> SELECT INSTANCE_NAME FROM V$INSTANCE
```

```
Command Prompt - sqlplus s  X + v  
Microsoft Windows [Version 10.0.22631.4037]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\Eric>sqlplus sys@ORACLE21CDB as sysdba  
  
SQL*Plus: Release 21.0.0.0.0 - Production on Sun Sep 22 13:29:50 2024  
Version 21.3.0.0.0  
  
Copyright (c) 1982, 2021, Oracle. All rights reserved.  
  
Enter password:  
  
Connected to:  
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production  
Version 21.3.0.0.0  
  
SQL> |
```

- The command `sqlplus sys@ORACLE21CDB as sysdba;` connects you to the Oracle database ORACLE21CDB using the SYS user with SYSDBA privileges. This allows you to perform administrative tasks on the container database (CDB) or any pluggable database (PDB) within it.

```
SQL> Sqlplus sys@ORACLE21CDB as sysdba;
```

Switch to the PDB & Create User & Grant Privileges

```
Command Prompt - sqlplus s X + v
SQL> SELECT instance_name FROM v$instance;

INSTANCE_NAME
-----
oracle21cdb

SQL> ALTER SESSION SET CONTAINER = PDB2;

Session altered.

SQL> CREATE USER plsqlauca IDENTIFIED BY password;

User created.

SQL> GRANT ALL PRIVILEGES TO plsqlauca;

Grant succeeded.

SQL> |
```

- The command switches your current session to the pluggable database (PDB) named PDB2. This allows you to execute commands and queries specifically within that PDB context.

```
SQL> ALTER SESSION SET CONTAINER = PDB2;
```



- The command creates a new user named plsqlauca in the Oracle database, with the specified password. After executing this command, you may need to grant the necessary privileges to the user to allow them to perform actions within the database.

```
SQL> CREATE USER plsqlauca IDENTIFIED BY password;
```

- The command GRANT ALL PRIVILEGES TO plsqlauca; grants all available privileges to the user plsqlauca in the Oracle database. This allows the user to perform any action on the database, including creating, modifying, and deleting objects.

```
SQL> GRANT ALL PRIVILEGES TO plsqlauca
```


Connecting to Oracle PDB through Oracle SQL Developer for the Newly Created User



New / Select Database Connection

Connection Name	Connection Details
AUCA@2024	sys@//localhost:...
employeeManage...	sys@//localhost:...
orade21c_datab...	sys@//localhost:...
oradeclass	oradeclass@//loc...
oradepdb	sys@//localhost:...

Name:

Database Type:

User Info Proxy User

Authentication Type:

Username:

Password:

Role:

☐ Save Password

Connection Type:

Details Advanced

Hostname:

Port:

☐ SID

☒ Service name

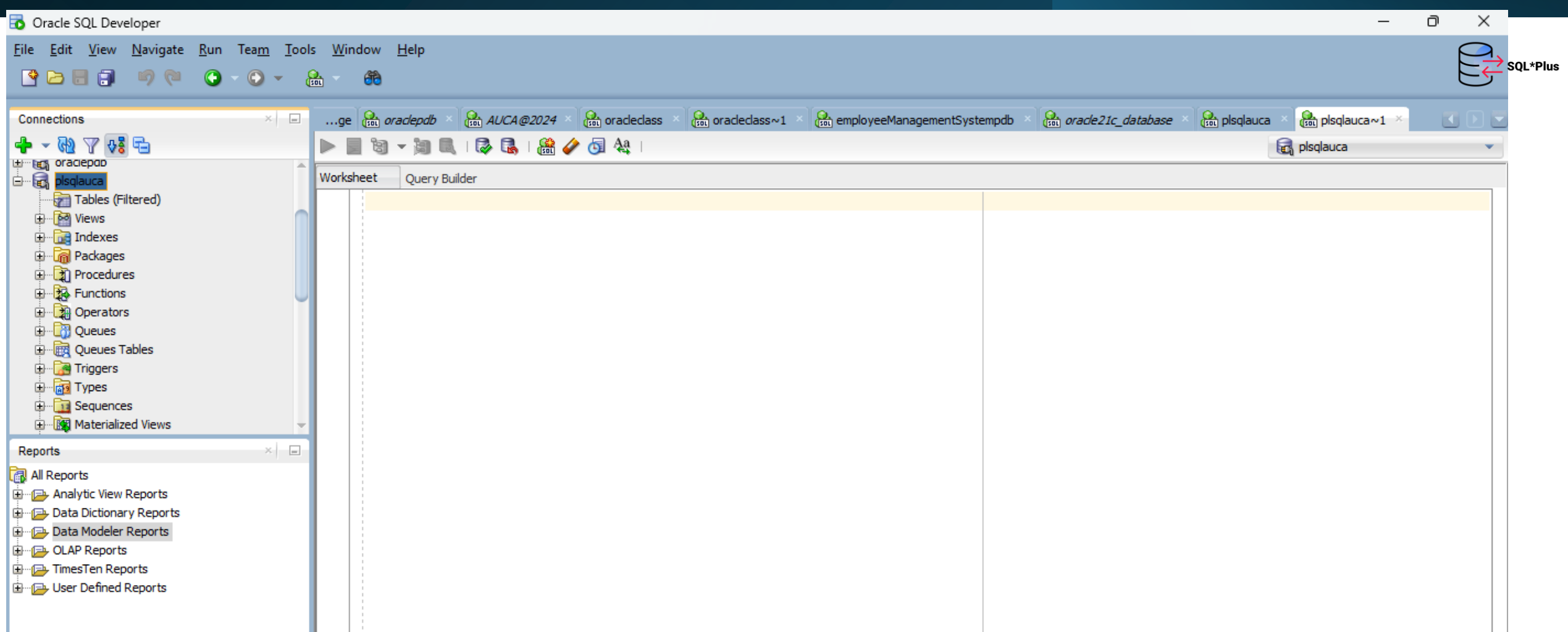
Status : Success

Oracle PL/SQL: User Creation and Privilege Grants

Command	Description
CREATE USER XXXXXX IDENTIFIED BY password;	Creates a new user with the given username (XXXXXX) and password.
GRANT ALL PRIVILEGES TO XXXXX	Granting all privileges to a user provides them with full access to the database.
GRANT CONNECT TO XXXXXX; GRANT CREATE SESSION TO XXXXXX;	Grants basic privileges for connecting to the database and creating a session.
GRANT DBA TO XXXXXX; GRANT RESOURCE TO XXXXXX; GRANT UNLIMITED TABLESPACE TO XXXXXX;	Grants DBA privileges, resource management, and unlimited tablespace usage.
GRANT CREATE VIEW TO XXXXXX; GRANT CREATE PROCEDURE TO XXXXXX; GRANT CREATE SEQUENCE TO XXXXXX; GRANT CREATE TRIGGER TO XXXXXX;	Allows the user to create views, procedures, sequences, and triggers.
GRANT ALTER ANY TABLE TO XXXXXX; GRANT ALTER ANY PROCEDURE TO XXXXXX; GRANT ALTER ANY TRIGGER TO XXXXXX;	Grants the ability to alter tables, procedures, and triggers.
GRANT DELETE ANY TABLE TO XXXXXX;	Allows the user to delete rows from any table in the database.
GRANT DROP ANY PROCEDURE TO XXXXXX; GRANT DROP ANY TRIGGER TO XXXXXX; GRANT DROP ANY VIEW TO XXXXXX;	Enables the user to drop (remove) procedures, triggers, and views.



Open worksheet for user: plsqaauca



Steps to Permanently Delete Pluggable Databases in Oracle



- **Close the PDBs:** Ensure the pluggable databases are not in use.
- **Unmount the PDBs:** Detach the PDBs from the container database.
- **Use the DROP PLUGGABLE DATABASE Command:** Execute with the INCLUDING DATAFILES option to remove both the PDBs and their associated data files.
- **Permanent Deletion:** This process permanently deletes the databases.



Step 1: Delete the pluggable databases HR and PDB4, follow these steps

```
Command Prompt - sqlplus s
C:\Users\Eric>sqlplus sys/admin@localhost:1521/ORACLE21CDB AS SYSDBA

SQL*Plus: Release 21.0.0.0.0 - Production on Wed Sep 25 15:51:22 2024
Version 21.3.0.0.0

Copyright (c) 1982, 2021, Oracle. All rights reserved.

Connected to:
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

SQL> col name for a20;
SQL> /
SP2-0103: Nothing in SQL buffer to run.
SQL>
SQL> select name, open_mode from v$containers
2 /

NAME                                OPEN_MODE
-----
CDB$ROOT                            READ WRITE
PDB$SEED                            READ ONLY
ORACLE21DBCPDB                      READ WRITE
HR                                  READ WRITE
PDB4                                READ WRITE

SQL> |
```

- To connect to an Oracle database using SQL*Plus as the SYS user with SYSDBA privileges.

C:\Users\Eric>sqlplus sys as SYSDBA

or

C:\Users\Eric> sqlplus sys/admin@localhost:1521/ORACLE21CDB AS SYSDBA

- The query retrieves the names and open modes of all pluggable databases in the current container database.

SQL> col name for a20; -- This is for formatting our output.

SQL> select name, open_mode from v\$containers

Step 2: Close and check the paths of Unplug DB

```
SQL> ALTER PLUGGABLE DATABASE HR CLOSE IMMEDIATE;

Pluggable database altered.

SQL> ALTER PLUGGABLE DATABASE PDB4 CLOSE IMMEDIATE;

Pluggable database altered.

SQL> |
```

- Before dropping a PDB, it needs to be closed.

```
SQL> ALTER PLUGGABLE DATABASE HR CLOSE IMMEDIATE;
SQL> ALTER PLUGGABLE DATABASE PDB4 CLOSE IMMEDIATE;
```

```
SQL> SELECT directory_name, directory_path FROM dba_directories;

DIRECTORY_NAME
-----
DIRECTORY_PATH
-----
DATA_PUMP_DIR
C:\oracle21c_config\admin\oracle21cdb\dpdump/

DBMS_OPTIM_LOGDIR
C:\oracle21c\cfgtoollogs

DBMS_OPTIM_ADMINDIR
C:\oracle21c\rdbms\admin
```

- Find or set the correct directory path for unplugging the PDBs.
- ```
SQL> SELECT directory_name, directory_path FROM dba_directories;
```

# Step 3: Use the Path in the UNPLUG Command



```
SQL> ALTER PLUGGABLE DATABASE HR UNPLUG INTO 'C:\oracle21c_config\admin\oracle21cdb\dpdump\hr.xml';
```

```
Pluggable database altered.
```

```
SQL> ALTER PLUGGABLE DATABASE PDB4 UNPLUG INTO 'C:\oracle21c_config\admin\oracle21cdb\dpdump\pdb4.xml';
```

```
Pluggable database altered.
```

```
SQL>
```

- Before dropping a PDB, it needs to be closed.

```
SQL> ALTER PLUGGABLE DATABASE HR UNPLUG INTO 'C:\oracle21c_config\admin\oracle21cdb\dpdump\hr.xml';
```

```
SQL> ALTER PLUGGABLE DATABASE PDB4 UNPLUG INTO 'C:\oracle21c_config\admin\oracle21cdb\dpdump\pdb4.xml';
```

# Step 4: Drop Pluggable Databases 'HR' and 'PDB4' and Verify Their Deletion"

```
SQL> DROP PLUGGABLE DATABASE HR INCLUDING DATAFILES;
Pluggable database dropped.

SQL> DROP PLUGGABLE DATABASE PDB4 INCLUDING DATAFILES;
Pluggable database dropped.

SQL>
SQL>
SQL>
SQL> SELECT name, open_mode FROM v$containers;

NAME OPEN_MODE
----- -
CDB$ROOT READ WRITE
PDB$SEED READ ONLY
ORACLE21DBCPDB READ WRITE
```

- Before dropping a PDB, it needs to be closed.

```
SQL> DROP PLUGGABLE DATABASE HR INCLUDING DATAFILES;
SQL> DROP PLUGGABLE DATABASE PDB4 INCLUDING DATAFILES;
```



- The query shows PDB (All pluggable database) names and their open modes.
- ```
SQL> SELECT name, open_mode FROM v$containers;
```

Notes:

- **CDB\$ROOT**: The root container of the multitenant database. It cannot be deleted.
- **PDB\$SEED**: A template PDB used for creating new PDBs. It cannot be deleted.
- **ORACLE21DBCPDB**: A user-created PDB that can be deleted, but ensure it's not in use and back up important data before proceeding.

Steps for Cloning a Pluggable Database (PDB)

In Oracle, **cloning** refers to creating an exact copy of a database or its components. This process is commonly used for tasks such as backup, testing, or creating development environments without affecting the production database. Cloning helps maintain data consistency while allowing operations on the copied database.



There are several types of cloning in Oracle:

- 1. Database Cloning:** Creating a duplicate of an entire Oracle database. This is useful for testing or performing operations without affecting the original database.
- 2. PDB (Pluggable Database) Cloning:** In Oracle Multitenant Architecture, you can clone a PDB within the same container database (CDB) or move it to a different CDB. It can be done as either:
 - **Cold cloning** (when the source PDB is closed)
 - **Hot cloning** (while the source PDB is open)
- 3. Schema Cloning:** Copying an individual schema from one database to another. This includes all objects within the schema, such as tables, indexes, views, and triggers.
- 4. Tablespace Cloning:** Duplication of a specific tablespace. This might involve creating a read-only copy of a tablespace for reporting purposes.

The cloning process can be done manually using commands like RMAN (Recovery Manager), Data Pump, or through Oracle Enterprise Manager, depending on the specific need.

Step 1: Logging as SYSDBA & Check available PDB

```
Command Prompt - sqlplus s
Microsoft Windows [Version 10.0.22631.4037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Eric>sqlplus sys/admin@localhost:1521/ORACLE21CDB AS SYSDBA

SQL*Plus: Release 21.0.0.0.0 - Production on Wed Sep 25 16:47:45 2024
Version 21.3.0.0.0

Copyright (c) 1982, 2021, Oracle. All rights reserved.

Connected to:
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

SQL> col name for a20;
SQL> select name, open_mode from v$containers
2 /

NAME                                OPEN_MODE
-----
CDB$ROOT                            READ WRITE
PDB$SEED                            READ ONLY
ORACLE21DBCPDB                      READ WRITE

SQL> |
```

- To connect to an Oracle database using SQL*Plus as the SYS user with SYSDBA privileges.

SQL> sqlplus sys as SYSDBA or

C:\Users\Eric> sqlplus sys/admin@localhost:1521/ORACLE21CDB AS SYSDBA

- The query retrieves the names and open modes of all pluggable databases in the current container database.

SQL> col name for a20; -- This is for formatting our output.

SQL> select name, open_mode from v\$containers

Step 2: Check if You Are on the Root Container and Display All PDBs

```
SQL>
SQL>
SQL> show con_name;

CON_NAME
-----
CDB$ROOT
SQL>
SQL> show pdbs;

  CON_ID CON_NAME              OPEN  MODE RESTRICTED
  -----
       2 PDB$SEED                READ ONLY        NO
       3 ORACLE21DBCPDB       READ WRITE       NO

SQL>
```

- This command will return the name of the current container you are connected to, which could be the root container (CDB\$ROOT) or a specific pluggable database (PDB).

SQL> Show con_name;

or

SQL> SELECT SYS_CONTEXT('USERENV', 'CON_NAME') FROM DUAL;

- Oracle instance that is currently running, which is usually the same as the **SID** (System Identifier) of the database.

SQL> SHQW PDBS;

Step3: Identify the right path of our PDB

```
SQL>
SQL> SELECT CON_ID, TABLESPACE_NAME, FILE_NAME FROM CDB_DATA_FILES WHERE CON_ID = 3;
```

CON_ID	TABLESPACE_NAME	FILE_NAME
3	SYSTEM	E:\ORACLE21CDATA\ORACLE21CDB\ORACLE21DBCPDB\SYSTEM01.DBF
3	SYSAUX	E:\ORACLE21CDATA\ORACLE21CDB\ORACLE21DBCPDB\SYSAUX01.DBF
3	UNDOTBS1	E:\ORACLE21CDATA\ORACLE21CDB\ORACLE21DBCPDB\UNDOTBS01.DBF

CON_ID	TABLESPACE_NAME	FILE_NAME
3	USERS	E:\ORACLE21CDATA\ORACLE21CDB\ORACLE21DBCPDB\USERS01.DBF

- Retrieves the tablespace names and data file locations for the PDB with CON_ID = 3. Based on the result, the **ORACLE21DBCPDB** (Pluggable Database 1) is located at the following file paths:

```
SQL> SELECT CON_ID, TABLESPACE_NAME, FILE_NAME FROM
CDB_DATA_FILES WHERE CON_ID = 3;
```

- This path is commonly used for organizing database files, making it easier to manage multiple PDBs within a CDB.

Step 4: Clone the PDB

```
SQL>
SQL> CREATE PLUGGABLE DATABASE plsql2024
2 FROM ORACLE21DBCPDB
3 FILE_NAME_CONVERT = ('E:\oracle21cdata\ORACLE21CDB\oracle21dbcpdb\' ,
4 'E:\oracle21cdata\ORACLE21CDB\oracle21dbcpdb\plsql2024\');
```

Pluggable database created.

```
SQL>
SQL>
SQL>
SQL> show pdbs;
```

CON_ID	CON_NAME	OPEN MODE	RESTRICTED
2	PDB\$SEED	READ ONLY	NO
3	ORACLE21DBCPDB	READ WRITE	NO
5	PLSQL2024	MOUNTED	

```
SQL>
SQL> ALTER PLUGGABLE DATABASE plsql2024 OPEN;
```

Pluggable database altered.

```
SQL> |
```

- Command to clone existing PDB "**ORACLE21DBCPDB**" to a new one called **plsql2024**

```
SQL> CREATE PLUGGABLE DATABASE plsql2024
FROM ORACLE21DBCPDB
FILE_NAME_CONVERT =
('E:\oracle21cdata\ORACLE21CDB\oracle21dbcpdb\' ,
'E:\oracle21cdata\ORACLE21CDB\oracle21dbcpdb\plsql2024\');
```

- Check existing PDB, plsql2024 is created with mounted status, which means that not yet opened.
- Open the new PDB (plsql2024)

```
SQL> show pdbs;
SQL> ALTER PLUGGABLE DATABASE plsql2024 OPEN;
```

Step 5: Save the State (optional) & Create User in the New PDB

```
SQL> show con_name;

CON_NAME
-----
CDB$ROOT
SQL>
SQL> ALTER SESSION SET CONTAINER=plsql2024;

Session altered.

SQL> CREATE USER adminpdb IDENTIFIED BY admin;

User created.

SQL> GRANT ALL PRIVILEGES TO adminpdb;

Grant succeeded.

SQL>
SQL>
SQL> SELECT username FROM dba_users WHERE username = 'ADMINPDB';

USERNAME
-----
ADMINPDB

SQL> |
```

- Save the state: Ensure the PDB opens automatically upon the next startup of the CDB:
SQL> ALTER PLUGGABLE DATABASE plsql2024 SAVE STATE;

- Shift from the root Container to the new DB
SQL> ALTER SESSION SET CONTAINER=plsql2024;

- Create a new user for our new Database plsql2024
SQL> CREATE USER adminpdb IDENTIFIED BY admin;

- Grant all privileges to the new user
SQL> GRANT ALL PRIVILEGES TO admin;
SQL> Display the user created

Step 6: Connect my new user to sql developer

The screenshot shows the Oracle SQL Developer interface with the 'New / Select Database Connection' dialog box open. The dialog is configured for a new connection named 'plsql2024_class'.

Connections Panel (Left):

- Oracle Connections
 - AUCA@2024
 - class2024
 - employeeManagementSystempdb
 - orade21c_database
 - oradeclass
 - oradepdb
 - plsqlauca
- Database Schema Service Connections

Dialog Box: New / Select Database Connection

Connection Name: plsql2024_class

Database Type: Oracle

User Info: Proxy User

Authentication Type: Default

Username: adminpdb

Role: default

Password: [Masked]

Connection Type: Basic

Details: Advanced

Hostname: localhost

Port: 1521

Service name: plsql2024

Status: Success

Buttons: Save, Clear, Test, Connect, Cancel

Footer:

- SQL Developer Forum
- Team Blogs and Magazine Articles
- SQLcl - The power of SQL Developer in a CLI
- Oracle Live SQL - Learn and share SQL, for free.

Note on Database User Management

- After configuring the database and creating a user with all privileges, you can interact with the databases effectively.
- Be cautious when granting privileges to users:
 - **Security Risks:** Excessive permissions can lead to unauthorized access.
 - **Data Integrity Issues:** Improper access can compromise data integrity.
- Follow the principle of least privilege:
 - Grant only the permissions necessary for users to perform their tasks.
- Regularly review user privileges:
 - Adjust permissions as needed to maintain a secure database environment.



Oracle Enterprise Manager

Oracle Enterprise Manager (OEM) is a comprehensive management tool **designed for monitoring, managing, and optimizing Oracle environments, including databases, middleware, and applications.** Here are some key features:



Monitoring

- OEM provides real-time monitoring of database performance, availability, and resource utilization.

Performance Tuning

- It enables administrators to perform tasks such as backup and recovery, patch management, and configuration management.

Reporting

- It includes reporting features to generate insights on performance metrics, user activities, and system health.

Centralized Control

- It provides a centralized interface to manage multiple Oracle instances across different environments, facilitating easier administration.

Overall, Oracle Enterprise Manager is essential for ensuring the health and efficiency of Oracle database environments.

How to open Oracle Enterprise Manager: Connect to Sys



```
Windows PowerShell
PS C:\Users\Eric> sqlplus sys as sysdba

SQL*Plus: Release 21.0.0.0.0 - Production on Sun Sep 22 22:32:16 2024
Version 21.3.0.0.0

Copyright (c) 1982, 2021, Oracle. All rights reserved.

Enter password:

Connected to:
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

SQL> show con_name;

CON_NAME
-----
CDB$ROOT
SQL> ALTER SESSION SET CONTAINER = CDB$ROOT;

Session altered.

SQL> SELECT SYS_CONTEXT('USERENV', 'CON_NAME') AS current_container FROM dual;

CURRENT_CONTAINER
-----
CDB$ROOT
```

In this:

- **Connect to the System**

```
SQL> sqlplus sys as sysdba
```

- **Check if I am connected to the root container**

```
SQL> show con_name
```

- If not, I should use:

```
SQL> ALTER SESSION SET CONTAINER = CDB$ROOT;
```

Or

- Check the current container

```
SQL> SELECT SYS_CONTEXT('USERENV', 'CON_NAME') AS
current_container FROM dual;
```

How to open Oracle Enterprise Manager: Verify configuration & set HTTP(s)

```
SQL> SELECT DBMS_XDB_CONFIG.GETHTTPPORT() AS HTTP_PORT,  
2          DBMS_XDB_CONFIG.GETHTTPS_PORT() AS HTTPS_PORT  
3 FROM dual;
```

HTTP_PORT	HTTPS_PORT
0	0

```
SQL>  
SQL> BEGIN  
2     DBMS_XDB_CONFIG.SETHTTPPORT(8080);  
3     DBMS_XDB_CONFIG.SETHTTPS_PORT(8443); -- Optional for HTTPS  
4 END;  
5 /
```

PL/SQL procedure successfully completed.

```
SQL>  
SQL>  
SQL> SELECT DBMS_XDB_CONFIG.GETHTTPPORT() AS HTTP_PORT,  
2          DBMS_XDB_CONFIG.GETHTTPS_PORT() AS HTTPS_PORT  
3 FROM dual;
```

HTTP_PORT	HTTPS_PORT
8080	8443

- **Verify Configuration:** After restarting, run the initial query again to confirm the new port settings.

- `SELECT DBMS_XDB_CONFIG.GETHTTPPORT() AS HTTP_PORT,
DBMS_XDB_CONFIG.GETHTTPS_PORT() AS HTTPS_PORT FROM dual;`

- Since both HTTP_PORT and HTTPS_PORT are returning 0, it confirms that the HTTP(S) server is currently not enabled. Here's how to proceed:
- Set New HTTP(S) Ports: Let's set the HTTP and HTTPS ports to a specific value. Choose ports that are not in use (for example, 8080 for HTTP and 8443 for HTTPS):

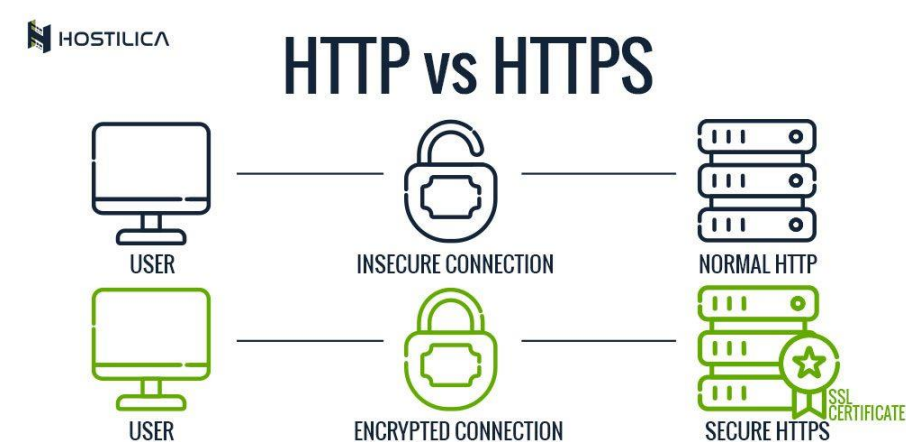
```
SQL> BEGIN  
    DBMS_XDB_CONFIG.SETHTTPPORT(8080);  
    DBMS_XDB_CONFIG.SETHTTPS_PORT(8443); -- Optional for HTTPS  
END;  
/
```

- Check for Errors: If you encounter a port conflict again, try different port numbers.
- Verify Configuration: After executing the above, check the port settings again:

```
SQL> SELECT DBMS_XDB_CONFIG.GETHTTPPORT() AS HTTP_PORT,  
          DBMS_XDB_CONFIG.GETHTTPS_PORT() AS HTTPS_PORT  
FROM dual;
```

Using LSNRCTL to Check Oracle Listener Status

```
Administrator: Command Prompt
C:\Windows\System32>lsnrctl status | findstr HOST
Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=Eric-Admin)(PORT=1521)))
(DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=Eric-Admin)(PORT=1521)))
(DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=Eric-Admin)(PORT=5500))(Presentation=HTTP)(Session=RAW))
(DESCRIPTION=(ADDRESS=(PROTOCOL=tcps)(HOST=Eric-Admin)(PORT=5501))(Security=(my_wallet_directory=C:\ORACLE21C_CONFIG\admin\oracle21cdb\xdb_wallet))(Presentation=HTTP)(Session=RAW))
(DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=Eric-Admin)(PORT=8080))(Presentation=HTTP)(Session=RAW))
(DESCRIPTION=(ADDRESS=(PROTOCOL=tcps)(HOST=Eric-Admin)(PORT=8443))(Security=(my_wallet_directory=C:\ORACLE21C_CONFIG\admin\oracle21cdb\xdb_wallet))(Presentation=HTTP)(Session=RAW))
C:\Windows\System32>
```



In this:

- **Identify the hostname that correspond to the port number 8443**

```
C:\Windows\System32>lsnrctl status | findstr HOST
```

How to open Oracle Enterprise Manager: Restart database

```
SQL> SHUTDOWN IMMEDIATE;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL> STARTUP;
ORACLE instance started.

Total System Global Area 2147482528 bytes
Fixed Size                  9857952 bytes
Variable Size             1392508928 bytes
Database Buffers          738197504 bytes
Redo Buffers               6918144 bytes
Database mounted.
Database opened.
SQL>
SQL> select dbms_xdb_config.gethttpsport() from dual;

DBMS_XDB_CONFIG.GETHTTPSPORT()
-----
                        8443

SQL> |
```

- Restart the Database: If the ports were successfully set, restart the database to apply changes:
SQL> SHUTDOWN IMMEDIATE;
STARTUP;

- This will display both port numbers in one result set.
SQL> select dbms_xdb_config.gethttpsport() from dual;

Connecting to Oracle PDB Locally Using SQL*Plus for the Newly Created User

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Eric> sqlplus plsqlauca/password@//localhost:1521/PDB2

SQL*Plus: Release 21.0.0.0.0 - Production on Sun Sep 22 21:09:27 2024
Version 21.3.0.0.0

Copyright (c) 1982, 2021, Oracle. All rights reserved.

Last Successful login time: Sun Sep 22 2024 19:15:47 +02:00

Connected to:
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

SQL> |
```



In this:

- **plsqlauca:** Username.
- **password:** Password for the plsqlauca user.
- **localhost:** Refers to your local machine.
- **1521:** The default Oracle listener port.
- **PDB2:** The service name of the pluggable database you're connecting to.

```
SQL> sqlplus plsqlauca/password@//localhost:1521/PDB2
```

How to open Oracle Enterprise Manager: Restart database

PL/SQL Control Statements x 10 Handling PL/SQL Errors x PL/SQL For Loop - Geeksfor x PL/SQL - Loops x Sign In To Oracle Enterprise x

Not secure https://localhost:8443/em/login

YouTube Maps Gmail Messaging Wix Website Editor |...



ORACLE ENTERPRISE MANAGER DATABASE EXPRESS

Username

Password

Container Name

- To access Oracle Enterprise Manager, open your web browser and enter the following URL:
<https://localhost:8443/em>

- The username should be "system" or "sys" as used during database creation.

- Use the password that was set during Oracle installation.
- Leave the container name blank.



Oracle Enterprise Manager – Tablespaces



ORACLE[®] Enterprise Manager Database Express

system ▾

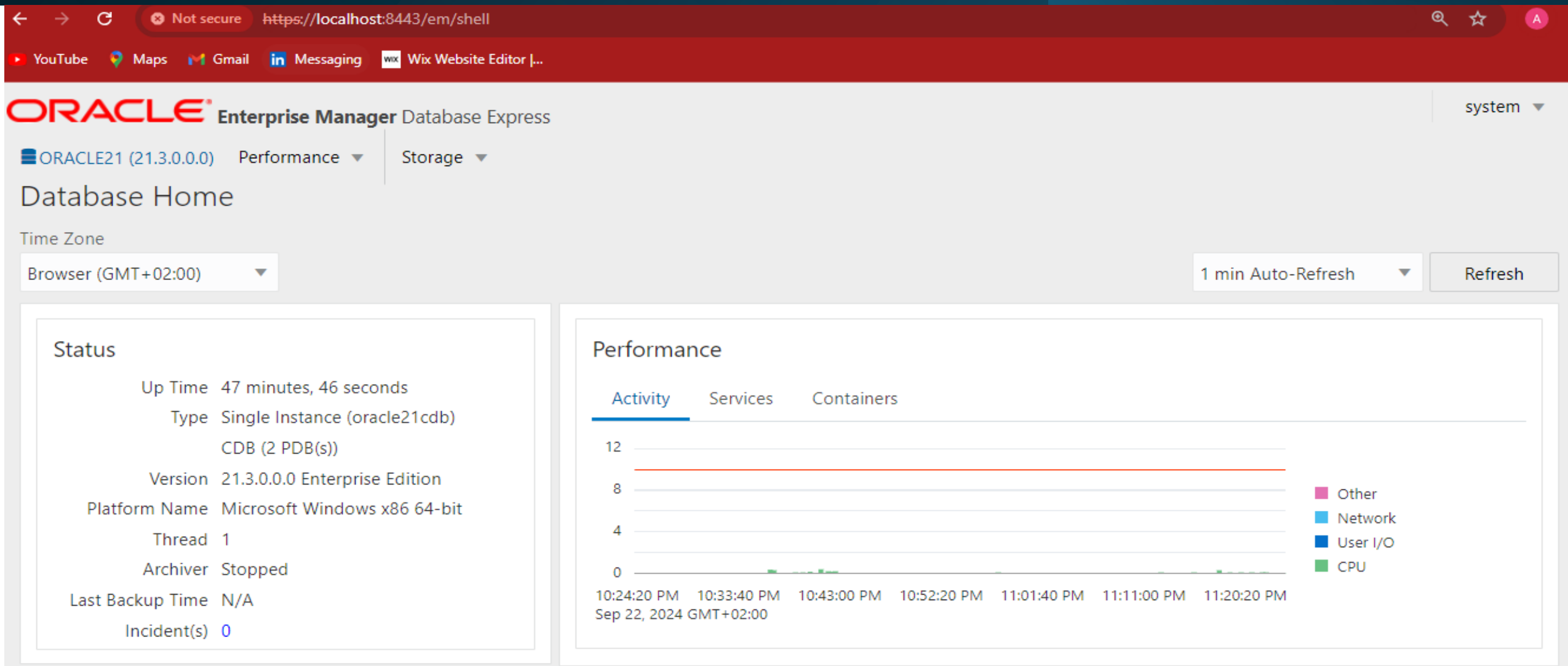
ORACLE21 (21.3.0.0.0) Performance ▾ Storage ▾

Tablespace

Filter by tablespace name

Name	Size	Used (%)	Auto Extend	Max Size	St...	Auto Segment Management	Directory
▶  SYSAUX	1.1 GB	92.3%	✓	UNLIMITED	●	✓	E:/ORACLE21CDATA/ORACLE21CDB/
▶  SYSTEM	1.3 GB	99.7%	✓	UNLIMITED	●		E:/ORACLE21CDATA/ORACLE21CDB/
▶  TEMP	237 MB	0.0%	✓	UNLIMITED	●		E:/ORACLE21CDATA/ORACLE21CDB/
▶  UNDOTBS1	110 MB	31.1%	✓	UNLIMITED	●		E:/ORACLE21CDATA/ORACLE21CDB/
▶  USERS	5 MB	53.8%	✓	UNLIMITED	●	✓	E:/ORACLE21CDATA/ORACLE21CDB/

Oracle Enterprise Manager – Database Home



It's time for you to explore Oracle Enterprise Manager (OEM) and get hands-on experience!



Simplified Administration: OEM offers a unified platform that makes database management intuitive and efficient.

Powerful Tools: You'll have access to robust features for monitoring, managing, and optimizing Oracle environments.

High Availability: OEM helps ensure that databases are always running at peak performance.

Streamlined Operations: Routine tasks are automated, reducing complexity and making your job easier.

Maximized Efficiency: Learning to use OEM will help you get the most out of any Oracle infrastructure.

Assignment II : Assignment: Database Creation, Deletion & EOM (1/2)

Task 1: Create a New Pluggable Database (PDB)

Each student is required to create a PDB named `plsql_class2024db`.

- **Username Format:** Your username should begin with the first two letters of your first name followed by `_plsqlauca`.
Example: If your first name is `Eric`, your username will be `er_plsqlauca`.
- **Password:** Choose a personal password, preferably something simple for ease of use.
- **Purpose:** This username will be used for class activities where you will store all your work during the course.



Task 2: Create and Delete a Pluggable Database

Instructions:

- You are required to create another Pluggable Database (PDB) based on the first two letters of your first name followed by `_to_delete_pdb`.
- After creating the PDB, you must delete it.

Example: If your first name is `Eric`, you will:

- Create a PDB named `er_to_delete_pdb`.
- Once the PDB is successfully created, proceed to delete it.

Ensure you take screenshots of both the creation and deletion process, and include all SQL commands used.

Task 3: Configure Oracle Enterprise Manager

Configure your Oracle Enterprise Manager and provide a screenshot of the dashboard after completing the above tasks.



Assignment: Database Creation, Deletion & EOM (2/2)

Task 4: Classroom Submission

- Create a GitHub repository for this assignment with a name related to the task.
- Ensure you take screenshots of each step (creating, deleting the PDB, and Oracle Enterprise Manager dashboard).
- Include all the queries you used in the README file of your repository, along with explanations.
- Add me as a collaborator on the repository.
- Note: Use the GitHub username **ericmaniraguha** or the email **ericmaniraguha@gmail.com** when adding me as a collaborator.
- Submit the link to your GitHub repository by filling out this [[Google Form](#)].



Grading and Deadlines

- This assignment is worth **10 points**.
- Late submissions will result in a **3-point deduction**.

Start Date: 26th September 2024

Due Date: 3rd October 2024, 11h59 pm

No excuses! Make sure to complete all tasks and submit on time.



Keep Learning!